

Section 2: Structures and functions in living organisms

- a) Levels of organisation
- b) Cell structure
- c) Biological molecules
- d) Movement of substances into and out of cells
- e) Nutrition
- f) Respiration
- g) Gas exchange
- h) Transport
- i) Excretion
- j) Coordination and response

a) Levels of organisation

Students will be assessed on their ability to:

- 2.1 describe the levels of organisation within organisms: organelles, cells, tissues, organs and systems.

b) Cell structure

Students will be assessed on their ability to:

- 2.2 recognise cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, chloroplast and vacuole
- 2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, chloroplast and vacuole
- 2.4 describe the differences between plant and animal cells.

c) Biological molecules

Students will be assessed on their ability to:

- 2.5 recall the chemical elements present in carbohydrates, proteins and lipids (fats and oils)
- 2.6 describe the structure of carbohydrates, proteins and lipids as large molecules made up from smaller basic units: starch and glycogen from simple sugar; protein from amino acids; lipid from fatty acids and glycerol
- 2.7 describe the tests for glucose and starch
- 2.8 understand the role of enzymes as biological catalysts in metabolic reactions
- 2.9 understand how the functioning of enzymes can be affected by changes in temperature
- 2.10 understand how the functioning of enzymes can be affected by changes in pH**
- 2.11 describe how to carry out simple controlled experiments to illustrate how enzyme activity can be affected by changes in temperature.

d) Movement of substances into and out of cells

Students will be assessed on their ability to:

- 2.12 recall simple definitions of diffusion, osmosis and active transport
- 2.13 understand that movement of substances into and out of cells can be by diffusion, osmosis and active transport